Wave function

The quantity whose variations make up matter wave is called wave function (Ψ).It describes state of a particle or system.it is a function of position coordinate and time. It is a complex function.Ψ itself has no physical interpretations. $/\Psi/^2$ represents probability density or probability of finding the particle in unit time. The problem of QM is to determine Ψ for a body when its freedom of motion is limited by the action of external forces.

If a particle exists in a given region of space, the total probability of finding the particle in that region is one. This idea can be put in mathematical form as

$$\Psi\ \Psi^*\ d\tau=1$$

dτ-volume element. It is called normalization condition.Ψ*-complex conjugate of Ψ.Complex conjugate of any function is obtained by replacing i by –i(imaginary part) wherever it appears in the function

$$\Psi= A+ i\ B$$

$$\Psi^*=A-iB$$

$$\Psi\ \Psi^*=A^2 - i^2B^2 = A^2+B^2$$

$$i^2=-1$$

Ψ Ψ* is called probability density. Ψ is not an observable quantity but Ψ Ψ* is an observable quantity.

Properties of the wave function

1. Wave function must be normalizable
2. Must be single valued
3. Must be finite everywhere
4. Must be continuous and should have a continuous first derivative everywhere

Ultrasonics

Frequencies greater than 20 KHz

Properties

1. They are highly energetic
2. When passed through the medium , at discontinuities they are partially reflected and this property is used in NDT
3. When absorbed by the medium, they generate heat.
4. They drill and cut thin metals
5. At room temperature, ultrasonic welding is possible.
6. Ultrasonic waves can be transmitted over long distances due to negligible loss of energy

Magnetostriction method

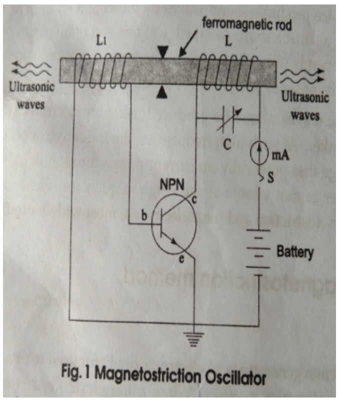
Principle:- when a magnetic field is applied parallel to the length of ferromagnetic rod made of material such as iron or nickel, a small elongation or contraction occurs in its length. The change in length depends on the intensity of the applied magnetic field and nature of ferromagnetic material. When the rod is placed inside a magnetic coil carrying alternating current, the rod suffers a change in length for each half cycle of alternating current. If the rod is made to vibrate with its natural frequency due to resonance the amplitude of vibration increases considerably. The natural frequency of vibration of rod f= (1/2l) (E/ρ)

l- length of the rod

E-Young's modulus of the material of the rod

ρ –density

if the frequency of the alternating current is sufficiently high, ultrasonic waves are produced from the ends of the rod.



An oscillator is a generator of alternating current. Generally used oscillator is Hartly oscillator. This oscillator is constructed with npn transistor. L-C circuit is connected to the collector. L1 is connected to base-emitter (input) and is kept close to L. the ferromagnetic rod is surrounded by L and L1. The inductors L and L1 and variable capacitor C together form a tank circuit. When oscillator is switched on, capacitor C gets charged. When it fully charged, it discharges through L and L1, setting up oscillations. In high frequency current, rod gets magnetized and demagnetized. Consequently length of rod changes and its free ends produce high frequency vibrations of ultrasonics. When frequency of oscillation produced by the oscillator equals natural frequency of rod, resonance occurs and amplitude of vibration increases considerably. The natural frequency of vibration of rod is given by
f= (1/2l) (E/ρ)

l- length of the rod,E-Young's modulus of the material of the rod, ρ –density

Advantages

1. Magnetostrictive materials are easily available and inexpensive
2. Oscillatory circuit is simple to construct
3. Large output power can be generated.

Disadvantages

1. Since the elastic constant varies with magnetization, one cannot get a single frequency oscillations.
2. Generation of frequencies beyond 300 KHz is not possible.
3. When a single rod of ferromagnetic material is used eddy current losses appear.

2. Piezoelectric method

When one pair of opposite faces of certain crystals such as quartz is subjected to pressure, in the other pair of opposite faces opposite electric charges are developed. When the faces are subjected to tension instead of pressure the sign of developed charges reverses. This effect is known as piezoelectric effect. The inverse piezoelectric effect is also true. According to this if alternating voltage is applied to one pair of faces, along the direction of the other pair of opposite faces the dimension varies ie, it expands and contracts periodically thereby generating elastic waves. When the frequency of alternating voltage matches with the natural frequency of the crystal,

$f=(1/2l)\ (E/\rho)$

$= \text{Ʊ}/2l$, resonant vibrations are produced resulting in generation of ultrasonic waves.

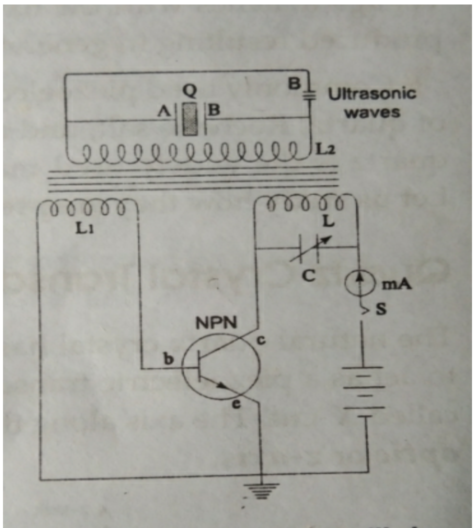


Fig. 3 Piezoelectric oscillator

Advantages

1. One can generate frequencies as high as 500 MHz
2. Stable single frequency waves can be generated
3. Using different transducers one can generate a wide range of frequencies

Disadvantages

Ferroelectric crystals are very expensive

Cutting and shaping are not easy.

Applications

Ultrasonic diffractometer, Industrial applications, Ultrasonic flow detector, Inspection techniques

Problems:-

1. Calculate the natural frequency of 40 mm length of a pure iron rod. Given that the density of pure iron is 7.25 x 10^3 kg/m^3. Young's modulus is 115 x 10^9 N/m^2 (ans:-49.78KHz)
2. Calculate the natural frequency of ultrasonic waves that can be generated by a nickel rod of length 4 cm. The density of nickel 8900 kg/m^3. Young's modulus of nickel 207GPa (1GPa=10^9 Pa) (ans:-60.28KHz)
3. Calculate the natural frequency of the fundamental note emitted by a piezoelectric crystal of vibrating length 3 cm. The density 2500 kg/m^3. Young's modulus 8 x 10^{10} N/m^2 (ans:-942.8 KHz)